

Electoral Equilibrium: Estimating Optimal Voter-Coalition Rebalancing under Political Shocks

[Working title — outline / draft scaffold]

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Abstract

Skeleton — fill after final runs. We present a three-stage framework that estimates how a party’s voter coalition must rebalance after a hypothetical political shock. A fine-tuned language model maps free-text shock descriptions to per-bloc loyalty shifts; a quasi-convex optimizer recovers the coalition weights that maximize win probability; and a compositional Monte Carlo returns a calibrated confidence interval. Our central methodological contribution is a *neutrality validation* protocol that separates structural (baseline) partisan bias from model-driven bias. [TODO: state headline numbers once final: structural asymmetry reduction $\sim 98\%$, residual base-model lean $+0.028$, final MAE.]

1 Introduction

Problem. Political shocks (scandals, endorsements, economic news, institutional events) re-sort the electorate: demographic blocs shift their support unevenly, so a campaign’s *optimal* coalition emphasis changes after a shock. We formalize this as recovering post-shock coalition weights that keep a party’s effective loyalty above an electoral-college-adjusted win threshold.

Data-scarcity gap. There are only ~ 20 U.S. presidential cycles (1948–2024), so the shock $\rightarrow$ response mapping cannot be learned by brute-force supervised regression on historical elections alone. We bridge this with (i) survey-derived demographic panels, (ii) social-media supervision, and (iii) synthetic shock–response data, and we push all small-sample uncertainty into an explicit covariance term rather than hiding it.

Research question. *Given a hypothetical shock, what is the optimal race-bloc coalition weighting, and with what probability does the resulting coalition win — and is that estimate partisan-neutral?*

Contributions.

- A constrained-decoding LLM stage that emits ordinal per-bloc loyalty shifts over 15 demographic blocs (5 race, 7 religion, 3 gender).
- A disciplined quasi-convex (DQCP) $\max P(\text{win})$ optimizer with demographic-plausibility bounds, plus a Logistic-Normal ILR Monte Carlo for calibrated win-probability confidence intervals.
- A neutrality-validation protocol (zero-shock null test + matched-pair symmetry test) that isolates and removes a structural baseline bias and attributes the residual lean to base-model priors.

2 Related Work

STUB — *literature not present in the repository*. [TODO: source: literature review not in repo. Cover: (a) demographic/coalition electoral modeling; (b) compositional data analysis (Aitchison) and portfolio optimization analogies; (c) LLMs as annotators / political bias in LLMs; (d) quasi-convex programming. Named anchors already in the repo: Diamond & Boyd 2019 (DQCP); Robinson 1950 (ecological fallacy); Ledoit–Wolf (shrinkage); Aitchison (ILR / compositional data).]

3 Data

3.1 Survey sources and panel construction

The pipeline uses **two distinct demographic inputs**, which must not be conflated:

- **Per-bloc loyalty μ — a multi-source survey merge.** Within-bloc Democratic vote shares are estimated by merging several independent surveys: the National Election Pool (NEP) exit polls, the ANES Cumulative Data File (1948–2024), the General Social Survey (GSS), and the Cooperative Election Study (CES, 2006–2024). Each survey is first aggregated to a bloc-level share *using its own sampling weights* (with Kish effective- N standard errors), and the sources are then reconciled per cycle**×**bloc cell by **inverse-standard-error weighting** (**resolve_conflicts**: the more precise survey counts more). The Democracy Fund VOTER Study Group panel and Pew NPORS are wired into the loader but *deferred* (wave-column parsing / party-proxy only), so they do not yet enter the merge.
- **Electorate composition w^0 — NEP exit-poll shares.** The baseline *size* of each race bloc, used to form the baseline coalition weights w^0 and the demographic-plausibility bounds in Stage 5, comes from **NEP exit-poll electorate shares** (White 0.63, Latino 0.15, African American 0.12, Asian 0.05, other 0.05), *not* from the survey merge. Loyalty μ answers “how Democratic is each bloc”; w^0 answers “how large is each bloc” — two separate quantities.

The panel is stored as three parallel, non-nested strata (race / religion / gender), each marginalizing to $\sim 100\%$ of the electorate. [TODO: source: confirm final panel row counts. Kernel run reported 305 rows $\rightarrow$ 220 after resolving 85 (cycle,bloc) conflicts; 14/15 blocs covered.]

3.2 Imputation and data-quality rules

- **1992 Perot three-party correction:** raw two-party shares are corrected using *bloc-specific* 1992 exit-poll Perot support (White 21%, African American 7%, Latino 14%), not a flat national adjustment.
- **ANES 2020 weights:** ANES is one of several merged loyalty sources; its 2020 sampling-weight column is entirely null, so that cycle falls back to equal-weighting, inflating some estimates (e.g. White Dem share ~ 3.6 pp above NEP). It is down-weighted / excluded from the affected race and gender merges — a real data-quality limitation, not silently corrected.
- **Thin-bloc imputation, later superseded:** early panel work carried forward sparse pre-1990 cells (e.g. **other_race** 1948–1956 carry-forward; an empty 1968 ANES **asian** cell). These are *excluded* by the current 1990+ covariance window (Sec. 4.6); do not present the earlier carry-forward as current.

3.3 Social-media supervision

Social posts provide RoBERTa sentiment-elasticity signal, tiered by how demographics are inferred, with strict rules on what may enter the covariance Σ_Δ :

- **user_bio** (Tier 1; Bluesky, X/Apify): SetFit bio classifier; included in both the mean μ and Σ_Δ .
- **subreddit_proxy / platform_proxy** (Reddit; TruthSocial, TikTok, 3DLNews): group/platform-level proxies; included in μ only, **excluded from** Σ_Δ (ecological-fallacy guard).
- **language_prior** (fallback): **excluded from both** μ and Σ_Δ (held-out only); mixing it would bifurcate the mean/covariance data sources and violate optimizer assumptions.

Sample-bias caveats. Social platforms over-represent the politically engaged and skew by platform (TruthSocial conservative; TikTok young/diverse); surveys under-represent low-response strata. These are carried as limitations, not silently corrected.

3.4 Synthetic shock-response data

Synthetic training records are generated by **DeepSeek** and reviewed by **Gemini** (dual-model to reduce single-model idiosyncrasy), over a balanced-seed taxonomy spanning event domain, ideological valence, electoral effect (the label target), and party perspective. An earlier, label-skewed version (Mistral v2 era) was **rebalanced** (to a -0.0007 label asymmetry) and the adapter retrained. The current merged corpus is **1,228** records (`data/finetune/synthetic.jsonl`), split **982** train / **246** eval (`data/finetune/train.jsonl`, `eval.jsonl`). [**TODO:** source: fill DeepSeek/Gemini model versions and the number of human-reviewed / directionally-validated examples (`docs/decisions.md:240` placeholder).] [**TODO:** source: MMD acceptance threshold not yet populated (`configs/mmd_config.json`).]

4 Methodology

4.1 Overview

Seven-stage pipeline: (1) voter panel $\rightarrow$ (2) baseline portfolio $\rightarrow$ (3a-d) sentiment / finetune / market $\rightarrow$ (4) LLM shock response $\rightarrow$ (5) DQCP optimization $\rightarrow$ (6) Monte Carlo. Deterministic under a single global seed (42).

4.2 Stage 4 — fine-tuned shock model

Base model `mistralai/Mistral-7B-v0.3`, QLoRA rank $r=16$, $\alpha=32$, dropout 0.05, target modules `{q_proj,k_proj,v_proj,o_proj}`, 4-bit NF4, lr 2×10^{-4} , 3 epochs, batch 4. The **primary** output is JSON over 15 blocs, each assigned one of **9 ordinal delta bins** (`strong_neg ... strong_pos`) tiling the clipped range $[-0.15, +0.15]$. A scalar `delta_eff` is a *secondary, derived* summary — the λ -weighted blend of the per-bloc deltas — not an independent prediction. **Constrained decoding** via `outlines` (Pydantic FSM) guarantees schema-valid tokens; greedy+regex is the fallback.

4.3 Stage 5 — DQCP max- $P(\text{win})$ optimizer

Decision variables are the 5 race-bloc weights w (simplex). Objective:

$$\max_w \Phi \left(\frac{\mu_{\text{eff}}(w) - V_{eq}}{\lambda_1 \sqrt{w^\top \Sigma_\Delta w}} \right), \quad \mu_{\text{eff}}(w) = \lambda_1 \sum_i w_i \tilde{\mu}_i^{\text{race}} + \lambda_2 \sum_R v_R \tilde{\mu}_R^{\text{rel}} + \lambda_3 \sum_G g_G \tilde{\mu}_G^{\text{gen}}.$$

Φ is the standard normal CDF (monotone), so this is a Sharpe-ratio form. **Quasiconvexity:** numerator affine, denominator convex-positive $\Rightarrow$ ratio quasi-convex (Diamond & Boyd 2019); solved as an SOCP via Charnes–Cooper (CLARABEL). **Bounds:** each bloc constrained to $[0.5 w_i^0, 1.5 w_i^0]$ around its baseline NEP share (demographic plausibility). **Win condition:** $\mu_{\text{eff}}(w) \geq V_{eq}$. **[TODO: source: include the formal quasiconvexity proof as an appendix (paper deliverable per DECISIONS.md).]**

4.4 Layer weights and win threshold

Layer weights $\lambda_{1,2,3}$: additive prior (0.5, 0.3, 0.2); IPF-raked posterior (0.114, 0.224, 0.662) — the paper compares additive vs raked. Win threshold V_{eq} : EC-adjusted logistic regression gives 0.5066 (Dem) / 0.4934 (Rep), with sensitivity bounds $[0.501, 0.515]$ (Dem) and $[0.485, 0.499]$ (Rep). Both the raked λ and V_{eq} are calibrated on the **full 1948–2024 panel (20 cycles)** — deliberately a wider window than the covariance Σ_Δ , which restricts to 1990+ (Sec. 4.6): the layer weights and win threshold want maximal historical leverage, whereas Σ_Δ excludes pre-1990 cycles to avoid thin-bloc sampling artifacts. **[TODO: source: report the V_{eq} logistic-regression coefficients / CI (derive_ec_veq in electoral/models/ml_baseline.py; 20-cycle fit, $C=10^4$).]**

4.5 Stage 6 — Logistic-Normal ILR Monte Carlo

Not Dirichlet (forbids the positive off-diagonal covariance of wave elections) and not the delta method. Steps: (1) ILR forward $z^* = V^\top \log w^*$ with Helmert contrast V ; (2) propagate $\Sigma_{\text{ILR}} = J \Sigma_\Delta J^\top$, $J = V^\top \text{diag}(1/w^*)$; (3) draw $y^{(n)} \sim \mathcal{N}(z^*, \Sigma_{\text{ILR}})$, $N=10,000$; (4) back-transform $w^{(n)} = \text{softmax}(V y^{(n)})$; (5) win fraction and 5th/95th-percentile CI. PSD repair floors eigenvalues at 10^{-6} .

4.6 Covariance Σ_Δ

Race-only 5×5 , estimated by **Ledoit–Wolf shrinkage**. Estimated over **9 cycles (1992–2024, 8 first-differences)**, restricted from the full 1948–2024 panel: pre-1990 cycles are excluded because thin blocs (`asian`, `latino`, `other_race`) had sparse/imputed sampling that inflated their variance $> 2 \times$ (e.g. `asian` 0.27 $\rightarrow$ 0.12). Shrinkage mitigates but does not eliminate the small-sample limitation for the thin blocs. **[TODO: source: report the Ledoit–Wolf shrinkage coefficient ρ as a transparency metric.]**

4.7 Second base model (controlled comparison)

A parallel path fine-tunes `google/gemma-2-9b` on the identical corpus and QLoRA config (bf16 + eager attention for Gemma 2 soft-capping), to test whether the residual lean (Sec. 5) is base-model-specific. **[TODO: source: Gemma 2 run not yet completed — no results.]**

5 Neutrality Validation

5.1 Zero-shock null test

Feeding all-zero deltas through the full equilibrium + simulation for both parties isolates *structural* baseline bias (independent of any LLM output). The pre-fix null was strongly asymmetric:

Party	μ_{eff}	margin	win prob.
Democrat	0.5575	+0.0509	1.00
Republican	0.5000	+0.0066	0.26

5.2 Root cause and fix

The bias was *not* in the ML model but in an **equal-within-stratum weighting placeholder** that over-weighted small, strongly-Democratic religion/ gender blocs (`jewish,muslim,other_gender`) by 7–33 $\times$. Replacing it with real electorate marginals moved the Democratic religion mean 0.543 $\rightarrow$ 0.499 and gender mean 0.567 $\rightarrow$ 0.505, cutting structural asymmetry +0.052 $\rightarrow$ +0.001 (**$\sim 98.2\%$ reduction**). The small residual slightly favors Republicans — the intended EC-adjusted-target effect.

5.3 Residual win-probability asymmetry

A residual win-probability asymmetry is *retained by design* as a coalition-*concentration* property (the optimizer concentrates emphasis differently per party), not a structural mean bias. **[TODO: source: state the exact retained win-prob asymmetry value + framing.]**

5.4 Matched-pair symmetry test and elimination

Matched party-flipped shock pairs reveal a residual behavioral lean of +0.028 (Democrat-favoring). Traced by elimination: (i) baseline aggregation — eliminated (Sec. 5.2); (ii) covariance artifacts — eliminated (Monte Carlo now consumes the real Σ_{Δ}); (iii) synthetic-corpus label imbalance — rebalanced to -0.0007 and the adapter retrained, yet the lean **persisted at** +0.028. **Conclusion:** the residual localizes to the base model’s pretrained political priors, which survive light LoRA fine-tuning, concentrating in valence-interpretation events. Disclosed as a limitation, not tuned away. **[TODO: source: full matched-pair table (per-pair asymmetries) from scripts/validate_symmetry.py final run.]**

6 Results

STUB — table skeletons; numbers pending final runs.

Metric		
Fine-tune held-out MAE (delta units)	0.0362 (<i>Mistral v2, pre-rebalance, 111-example eval</i>)	[TODO: ...]
Worst-bloc MAE (<code>other_gender</code>)		0.0488
Structural asymmetry (pre $\rightarrow$ post fix)		+0.
Residual base-model lean		
Base-model comparison (Mistral vs Gemma 2)		[TODO: ...]
Win-prob CI coverage / calibration		[TODO: ...]
Symmetry: mean magnitude ratio / asymmetry		[TODO: source: ...]

[TODO: source: worked end-to-end example (one shock $\rightarrow$ deltas $\rightarrow$ optimal weights $\rightarrow$ win-prob CI) as the results centerpiece.] **[TODO: source: latency numbers if reporting system performance (benchmark notes: mean ~ 11.82 s/inference).]**

7 Limitations

- **Additive independence / ecological fallacy** (Robinson 1950): the three strata are combined additively with no cross-tabulation; μ and V_{eq} coincide only under this approximation.
- **Model-internal win threshold:** V_{eq} is an EC-adjusted effective-loyalty threshold, not a literal popular-vote count.
- **Residual base-model lean** (+0.028): survives corpus rebalancing; disclosed rather than tuned away.
- **Synthetic-label provenance:** labels are LLM-generated (DeepSeek+Gemini); external-API use is a documented deviation from the reproducibility rule for this stage.
- **Race-only rebalancing:** only 5 race blocs are optimized; religion/ gender are fixed weights, and the Monte Carlo treats them as neutral ($\mu=0.50$), understating uncertainty.
- **Small-sample covariance:** 8 first-differences even after shrinkage; thin blocs remain data-limited.

8 Future Work

- IPF raking marginals to replace fixed religion/gender weights in the live path.
- Heavier fine-tuning (higher rank, more epochs, larger/curated corpus) to test whether the base-model lean can be reduced without over-fitting.
- Completing the Gemma 2 9B controlled comparison.
- Covariance sensitivity analysis (window, shrinkage target, estimator).
- Post-hoc calibration against prediction markets (benchmarks only).

9 References

STUB — bibliography. [TODO: source: build .bib. Anchors named in repo:]

- Diamond & Boyd (2019), *Disciplined Quasi-Convex Programming*, arXiv:1905.00562. [TODO: verify cite]
- Robinson (1950), ecological fallacy. [TODO: full cite]
- Ledoit & Wolf, covariance shrinkage. [TODO: year/venue]
- Aitchison, ILR / compositional data analysis. [TODO: full cite]
- Kuriwaki (2025), Cumulative CES Common Content, doi:10.7910/DVN/II2DB6.
- Foster & Rathlin (2022), MeToo sentiment dataset, DOI 10.5683/SP3/1YWCW1.
- [TODO: ANES, CES, GSS, NPORS, NEP, VOTER Panel dataset citations.]